

CyberSOC Platform

Admin Interface Specification

Complete Administrative Control Plane Design
Full Actions | RBAC | Tenant Management | System Operations

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1. Introduction & Overview

The CyberSOC Admin Interface serves as the central control plane for all platform management operations, providing authorized administrators with comprehensive capabilities to configure, monitor, and maintain the security operations platform. This specification defines the complete functional requirements, architectural design, user experience patterns, and integration points necessary for implementing a production-grade administrative interface that supports the full spectrum of operational activities from initial system configuration through ongoing management of multi-tenant deployments at enterprise scale.

The Admin Interface addresses a critical gap identified during the Go-Live readiness assessment: without centralized administrative controls, operational teams lack self-service capabilities for routine management tasks, creating dependency on engineering resources for activities that should be operator-executable. This specification establishes the foundation for empowering administrators while maintaining security through role-based access controls, audit logging of all administrative actions, and workflow approvals for high-impact operations that could affect platform stability or customer data.

1.1 Scope and Objectives

- **User Management:** Complete lifecycle management for all user accounts including creation, modification, deactivation, and deletion with appropriate approval workflows for sensitive operations
- **Role-Based Access Control:** Granular permission management supporting hierarchical roles, custom role definition, and attribute-based access control policies for complex authorization scenarios
- **Tenant Administration:** Multi-tenant management capabilities enabling MSSP operators to create, configure, and isolate customer tenants with resource quotas and white-label customization
- **System Configuration:** Centralized management of platform settings including detection rules, integration endpoints, notification channels, and feature flags controlling module availability
- **Licensing Operations:** License key management, entitlement viewing, usage monitoring, and integration with billing systems for financial operations
- **Audit & Compliance:** Comprehensive audit logging, compliance reporting, and evidence export supporting regulatory requirements and internal governance processes

1.2 Target User Personas

Persona	Description	Primary Needs	Access Level
Super Administrator	Platform owner with full system access	System config, tenant mgmt, billing	Full Access
Organization Admin	Manages users within their org	User mgmt, role assignment, reports	Org-scoped
Billing Administrator	Handles financial operations	Invoices, payments, credits	Financial scope
Security Administrator	Manages security configurations	Detection rules, integrations, alerts	Security scope
Read-Only Observer	Views dashboards and reports	Monitoring, auditing, read-only access	View Only

Table 1.1: Admin Interface User Personas

2. Architecture Design

The Admin Interface architecture follows a modern web application pattern separating presentation logic from backend services through well-defined APIs. The frontend implements a single-page application using React or Vue.js framework with component-based UI construction enabling consistent interaction patterns across administrative modules. Backend services expose RESTful APIs following OpenAPI 3.0 specification with authentication via JWT tokens issued after OAuth 2.0/OIDC authentication flow. All administrative actions generate audit events captured in immutable log storage supporting compliance investigations and operational review.

2.1 Component Architecture

Layer	Components	Technology	Responsibility
Presentation	Admin SPA, Component Library	React/Vue + TailwindCSS	UI rendering, user interactions
State Management	Redux/Pinia Store	Redux Toolkit / Pinia	Client state, caching, optimistic updates
API Gateway	Kong / Ambassador	Kubernetes Ingress	Auth, rate limiting, routing
Admin API Services	User/Role/Tenant/Config APIs	Go / Python FastAPI	Business logic, validation
Authorization Engine	OPA / Casbin	Open Policy Agent	Policy evaluation, ABAC
Data Layer	PostgreSQL + Redis	Relational + Cache	Persistent data, sessions
Audit Service	Event Logger	Apache Kafka + S3	Immutable audit trail
Integration Bus	Webhook/Event Emitter	NATS / Redis Streams	Async notifications

Table 2.1: Admin Interface Architecture Layers

2.2 Navigation Structure

Module	Sections	Key Pages	Icon
Dashboard	Overview, Quick Actions, Recent Activity	Main Dashboard, Widgets Config	grid-view
Users	List, Create, Edit, Bulk Import	User Profile, Permissions, Activity	users
Roles & Permissions	Roles List, Role Editor, Permission Matrix	Role Detail, Custom Policies	shield
Tenants	Tenant List, Tenant Editor, Quotas	Tenant Portal Preview, White-label	building
Configuration	General, Security, Integrations, Features	Setting Groups, Test Connections	settings
Licensing	Overview, Keys, Usage, Billing Sync	License Detail, Entitlement View	key
Integrations	Catalog, Active, Configuration	Integration Health, Logs	puzzle-piece
Audit Log	Search, Export, Attestation	Event Detail, Session Replay	clipboard-list
Monitoring	Health, Metrics, Alerts Status	Service Dashboard, Incident View	activity
Help & Support	Documentation, API Explorer, Tickets	KB Search, Support Chat	help-circle

Table 2.2: Admin Navigation Structure

3. User Management Module

The User Management module provides complete lifecycle administration for all platform user accounts. This includes creating new accounts with appropriate default configurations, modifying existing account attributes such as role assignments and contact information, managing authentication factors and credentials, handling account status transitions (activation, suspension, deactivation), and processing permanent deletions with appropriate data handling for retention requirements. The module supports both individual operations through form-based interfaces and bulk operations through CSV import/export for efficient large-scale administration.

3.1 User CRUD Operations

Operation	Description	Validation	Approval	Audit Level
Create User	Provision new user account	Email unique, role valid, quota checked	Auto/Optional	High
Read User	View user details and activity	Access scope verification	None	Standard
Update User	Modify user attributes	Field-specific rules	Sensitive fields	Medium
Deactivate User	Disable login access	No active sessions critical	Required	Critical
Delete User	Permanent removal	Data retention check, no assignments	Required + Delay	Critical
Bulk Import	CSV upload multiple users	Row validation, duplicate check	Batch approval	High
Bulk Export	Download user list	Scope filtering, PII masking	None	Standard
Password Reset	Force password change	MFA verification	Self/Admin	High

Table 3.1: User CRUD Operations Matrix

3.2 User Profile Schema

Field	Type	Required	Editable By	Notes
user_id	UUID	System	Read-only	Primary identifier
email	Email	Yes	Admin/Self	Unique, verified
display_name	String(100)	Yes	Admin/Self	Shown in UI
role_id	FK	Yes	Admin only	Determines permissions
tenant_id	FK	Yes	Super Admin	Multi-tenancy scope
status	Enum	System	Admin only	active/suspended/deleted
mfa_enabled	Boolean	No	Self/Admin	Security requirement
last_login	Timestamp	System	Read-only	Activity tracking
created_at	Timestamp	System	Read-only	Audit trail
custom_attrs	JSON	No	Admin	Extensible metadata

Table 3.2: User Profile Data Schema

3.3 Bulk Operations

Bulk operations enable efficient administration at scale, particularly valuable during initial tenant setup, organizational restructuring, or periodic maintenance activities. The bulk import function accepts CSV files with validated headers, performing row-by-row validation before committing changes. A preview mode shows anticipated results including any warnings or errors before final

confirmation. Bulk export supports filtering by various criteria (role, status, last activity date) and includes options for PII redaction when exporting for analysis or backup purposes. Both operations generate detailed audit records capturing the initiating user, timestamp, affected record counts, and individual row outcomes for traceability.

4. Role-Based Access Control Module

The RBAC module implements granular permission management enabling administrators to define roles containing specific capability sets, assign roles to users, and evaluate authorization requests consistently across all platform components. The implementation supports hierarchical role inheritance where child roles automatically acquire parent role permissions, custom role creation for organization-specific needs, and attribute-based access control (ABAC) policies addressing complex authorization scenarios requiring context-aware decisions based on environmental factors such as time of day, location, or device trust level.

4.1 Built-in Role Hierarchy

Role	Level	Scope	Capabilities	Limitations
super_admin	0 (Highest)	Global	All operations	None
org_admin	1	Organization	Org users, settings, reports	Cannot modify system config
billing_admin	1	Financial	Invoices, credits, payment methods	Cannot access user data
security_admin	1	Security	Rules, integrations, alerts	Cannot manage users
analyst_manager	2	Team	Team assignments, case mgmt	Cannot manage org settings
senior_analyst	3	Personal+	Investigate, respond, escalate	Cannot manage other users
analyst	4 (Base)	Personal	View assigned, basic actions	Limited configuration
readonly_viewer	5 (Lowest)	Assigned	View only, export reports	No write operations

Table 4.1: Built-in Role Hierarchy

4.2 Permission Model

Resource	Create	Read	Update	Delete	Special
users	admin	admin,self	admin,self	admin	password_reset, deactivate
roles	super_admin	admin	super_admin	super_admin	permission_grant
tenants	super_admin	admin	admin	super_admin	quota_adjust, suspend
config	super_admin	-	super_admin	-	feature_flags
licenses	billing_admin	billing_admin	billing_admin	-	credit_issue, refund
audit_log	-	admin	-	-	export, attest
integrations	security_admin	security_admin	security_admin	security_admin	test_connection
reports	-	admin+	self	-	schedule, share

Table 4.2: Permission Matrix by Resource

4.3 Custom Role Creation

Organizations requiring specialized permission sets beyond built-in roles can create custom roles with precise capability selection. The role editor presents available permissions organized by resource category with checkbox selection for each operation type (Create, Read, Update, Delete, Special). Custom roles can optionally inherit from existing roles as a base, adding or removing specific capabilities as needed. Role naming conventions enforce uniqueness within organizational scope and prevent confusion with built-in roles. Changes to custom role definitions take effect immediately for

assigned users, with optional notification to affected individuals when significant permission changes occur.

5. Tenant Management Module (MSSP)

The Tenant Management module provides Multi-Tenant administration capabilities essential for Managed Security Service Provider (MSSP) deployments where a single CyberSOC instance serves multiple customer organizations. Each tenant maintains logical isolation of data, configurations, and user access while sharing underlying infrastructure for operational efficiency. The module supports the complete tenant lifecycle from initial provisioning through configuration, quota management, health monitoring, and eventual decommissioning, with white-label options enabling customer-branded portal experiences.

5.1 Tenant Lifecycle States

State	Description	Allowed Transitions	Data Handling	Billing Impact
Draft	Initial configuration in progress	Active, Archived	Isolated, mutable	Not billed
Active	Fully operational tenant	Suspended, Archived	Full access, audited	Active billing
Suspended	Temporarily disabled (non-payment)	Active, Archived	Read-only export	Suspended billing
Archived	Decommissioned but retained	Deleted (after retention)	Frozen, encrypted	Final invoice
Deleted	Permanently removed	Terminal state	Purged per policy	Credit if applicable

Table 5.1: Tenant Lifecycle State Machine

5.2 Tenant Configuration Options

Category	Setting	Options	Default	Tier Requirement
Branding	Logo Upload	Image (PNG, SVG, max 200KB)	CyberSOC logo	Business+
Branding	Primary Color	Hex color picker	#6f6751 (cascade)	Business+
Branding	Portal Name	String (3-50 chars)	Organization name	Business+
Branding	Custom Domain	CNAME configuration	*.cybersoc.cloud	Enterprise
Data	Retention Period	7/30/90/365/unlimited days	30 days	Varies by tier
Data	Daily Ingestion Limit	GB per day	Tier base amount	License-defined
Data	User Seat Limit	Maximum concurrent users	Tier base amount	License-defined
Features	Module Enable/Disable	Feature flag toggles	All enabled	License-defined
Features	AI Copilot Access	Enabled/Limited/Disabled	By tier	Professional+
Security	IP Allowlist	CIDR ranges	Open (0.0.0.0/0)	All tiers
Security	MFA Requirement	Optional/Required/Enforced	Optional	Configurable

Table 5.2: Tenant Configuration Options

5.3 Resource Quota Management

Resource quotas prevent any single tenant from consuming disproportionate platform resources, ensuring fair allocation and predictable costs for MSSP operators. Quotas apply to compute resources (CPU, memory, storage), data volumes (daily ingestion, retained data size, API call frequency), and operational limits (concurrent user sessions, SOAR actions per period, AI query volume). Administrators view current utilization against quotas via dashboard gauges with color-coded

indicators (green/yellow/red) approaching thresholds. Automated alerts notify operators when tenants exceed configurable threshold percentages (default 80% warning, 95% critical).

6. System Configuration Module

The System Configuration module centralizes management of all platform-wide settings affecting behavior, appearance, and integration behavior. Configuration is organized into logical groups with appropriate access controls restricting modification to authorized roles. Changes to production-affecting settings require confirmation dialogs explaining potential impact, with high-risk modifications optionally requiring secondary approval before taking effect. A configuration version history enables rollback to previous states if changes cause unexpected issues, with automatic snapshot capture before each modification.

6.1 Configuration Categories

Category	Settings Count	Access Role	Change Impact	Approval Required
General Platform	~25	super_admin	Platform-wide	High-risk items
Authentication & SSO	~15	super_admin, security_admin	User access	Yes (SSO changes)
Detection Rules	~50+	security_admin	Alert generation	Yes (production)
Notification Channels	~20	org_admin+	Alert delivery	No
Integration Endpoints	~30	security_admin	Data flows	Yes (credentials)
Feature Flags	~40	super_admin	Capability availability	Yes (major)
UI Customization	~15	org_admin	Appearance only	No
Backup & Retention	~10	super_admin	Data protection	Yes (retention decrease)
API Settings	~12	super_admin	Developer access	Yes (rate limits)

Table 6.1: Configuration Categories Overview

6.2 Feature Flag Management

Feature flags provide runtime control over platform capability availability without code deployment, enabling gradual rollouts, A/B testing, and rapid incident response through feature disabling. The admin interface exposes flag management with toggle controls, targeting rules (enable for specific tenants, user segments, or percentage rollout), and scheduling for future activation/deactivation. Each flag maintains change history showing who modified it, when, and what value was set, supporting rollback to previous states. Integration with the licensing system ensures flags respect entitled capabilities, preventing unauthorized feature access through flag manipulation alone.

Flag Name	Module	Type	Default	Rollout Strategy
ai_copilot_enabled	AI Engine	Boolean	By tier	Licensed tier gates
advanced_analytics	Dashboard	Boolean	Enterprise+	Progressive rollout
new_detection_ui	SIEM	Boolean	False	Canary → Percentage → Full
soar_v2_engine	SOAR	Boolean	False	Opt-in beta
threat_hunt_workspace	Threat Hunting	Boolean	Business+	Feature-gated
purple_team_mode	Platform	Boolean	False	Admin opt-in
api_rate_limit_v2	API	Integer	1000/min	Gradual increase

Flag Name	Module	Type	Default	Rollout Strategy
export_encryption	Reports	Boolean	True	Mandatory (GDPR)

Table 6.2: Sample Feature Flags

7. Licensing Operations Module

The Licensing Operations module provides administrative interfaces for managing license keys, viewing entitlement details, monitoring consumption against limits, and synchronizing with external billing systems. This module integrates closely with the Billing & Licensing Architecture defined in the companion specification document, exposing license management capabilities through the unified admin interface rather than requiring separate tools or direct database access. Role restrictions ensure sensitive licensing operations (key issuance, credit adjustments, refunds) are limited to appropriately authorized personnel with full audit trails.

7.1 License Key Management

Operation	Description	Available To	Workflow	Audit
View Licenses	List all license keys with status	Billing+, Super	Direct access	Standard
Issue License	Generate new license key	Super Admin only	Form → Review → Issue	Critical
Revoke License	Disable active license	Super Admin only	Select → Reason → Confirm	Critical
Extend License	Modify expiration date	Billing Admin	Request → Approval → Apply	High
Modify Entitlements	Change feature/user limits	Super Admin	Form → Validation → Apply	Critical
Transfer License	Move between tenants	Super Admin	Source → Dest → Confirm	High
Export Keys	Download license inventory	Billing Admin	Filter → Format → Download	Standard

Table 7.1: License Key Operations

7.2 Usage Monitoring Dashboard

The Usage Monitoring dashboard provides real-time visibility into consumption patterns across all licensed dimensions: user seat utilization, data ingestion volumes, API call frequencies, AI query counts, and SOAR action executions. Visualizations include time-series charts showing trends, gauge indicators displaying current consumption against limits as percentages, and tables ranking tenants or users by consumption volume. Automated alerts trigger when consumption approaches thresholds (configurable defaults at 80% warning, 95% critical), enabling proactive outreach before hard limits cause service disruption. Historical trending supports capacity planning and identifies growth patterns informing renewal discussions.

8. Audit Logging & Compliance Module

Every action performed through the Admin Interface generates immutable audit events captured in tamper-evident storage. The Audit Logging module provides search, filter, and export capabilities for investigating incidents, demonstrating compliance, and identifying anomalous administrative patterns. Events capture actor identity, action type, target resource, timestamp, source IP address, outcome (success/failure), and before/after values for modifications. Cryptographic chaining between events enables detection of retrospective tampering attempts, while append-only storage design prevents event deletion or modification after creation.

8.1 Audit Event Schema

Field	Type	Description	Example Value
event_id	UUID (v4)	Unique event identifier	550e8400-e29b-...
timestamp	ISO 8601	Event occurrence time	2025-01-15T10:23:45Z
actor_id	UUID	Who performed action	user_abc123...
actor_type	Enum	Actor classification	user / service / system
action	String	Operation performed	user.create / role.update
resource_type	String	Target entity category	user / role / tenant / config
resource_id	UUID	Specific target identifier	target_xyz789...
outcome	Enum	Result of operation	success / failure / denied
source_ip	IPv4/IPv6	Origin network address	203.0.113.42
changes	JSON	Before/after diff	{"status":["active","suspended"]}
reason	Text	Optional justification	Account deactivation request
signature	Bytes	Cryptographic integrity	sha256:abcdef...

Table 8.1: Audit Event Data Schema

8.2 Compliance Reporting

Report Type	Purpose	Content	Format	Frequency
Admin Activity Summary	Management oversight	All admin actions, counts by type	PDF, Dashboard	Weekly/Monthly
User Access Review	Access governance	User-role assignments, last login	Spreadsheet, PDF	Monthly (required)
Configuration Changes	Change management	All setting modifications with diffs	PDF, JSON	On-demand/Daily
Data Access Log	Privacy compliance	PII access events, exports	Encrypted CSV	On-demand (GDPR)
Security Incident Timeline	Incident response	Related events chronologically	PDF, Timeline view	On-demand
Attestation Package	Audit support	Evidence bundle for auditor	Signed PDF	Quarterly/Annual
Tenant Isolation Report	MSSP compliance	Cross-tenant access attempts	PDF	Monthly

Table 8.2: Compliance Report Types

9. Integration with Existing Platform Modules

The Admin Interface integrates with existing CyberSOC modules defined in the platform specification, extending administrative control to previously specified capabilities. This section documents key integration points ensuring the Admin Interface properly leverages and manages existing platform components rather than duplicating functionality. Integration follows established API contracts defined in Section 66 (API Documentation) with authentication flowing through the OAuth/OIDC infrastructure specified throughout the security architecture.

Platform Module	Spec Section	Admin Integration	API Dependencies
SIEM Engine	Sec 8-12	Detection rule management, log source configuration	/api/v1/siem/*
XDR Architecture	Sec 13-15	Endpoint agent deployment, policy push	/api/v1/xdr/*
UEBA Engine	Sec 16-18	Behavior model tuning, risk scoring parameters	/api/v1/ueba/*
SOAR Engine	Sec 19-21	Playbook management, action approval queue	/api/v1/soar/*
Threat Intel	Sec 22-24	Feed subscription, IOC source config	/api/v1/ti/*
AI SOC Copilot	Sec 46-51	Model access controls, query limits	/api/v1/ai/*
MSSP Platform	Sec 59	Tenant lifecycle, white-label, portals	/api/v1/mssp/*
Monitoring	Sec 76	Health checks, alert thresholds, dashboard	/api/v1/monitor/*
API Gateway	Sec 66	Key management, rate limit policies	/api/v1/admin/api-keys

Table 9.1: Platform Module Integration Points

9.1 Dashboard Integration

The Admin Interface provides dedicated dashboard views complementing the analyst, manager, and CISO dashboards specified in Section 61. While those dashboards focus on security operations content, the Admin Dashboard emphasizes operational health metrics: system resource utilization, active user counts, recent administrative actions, approaching quota thresholds, integration health status, and upcoming license expirations. Widget configuration allows administrators to customize their dashboard view prioritizing information relevant to their responsibilities. Cross-linking enables navigation from admin dashboards to relevant detail pages (clicking a user count widget navigates to User Management filtered appropriately).

10. Implementation Plan

Implementing the Admin Interface follows an incremental approach delivering value progressively while managing complexity. The plan organizes work into sprints focused on cohesive capability groups, with each sprint producing deployable increments that can be validated independently. Early sprints establish foundational infrastructure (framework, auth integration, navigation) enabling subsequent sprints to focus on domain-specific functionality. The timeline assumes a team of 2-3 frontend developers and 1-2 backend developers working in parallel with close collaboration on API contracts.

10.1 Sprint Plan

Sprint	Duration	Deliverables	Dependencies	Acceptance Criteria
Sprint 0: Foundation	1 week	Project scaffold, CI setup, design tokens	None	Build pipeline green
Sprint 1: Shell & Auth	2 weeks	App shell, nav, login, role-based routing	Sprint 0	Authenticated access works
Sprint 2: User Mgmt	2 weeks	User CRUD, profile editing, bulk ops	Sprint 1	Full user lifecycle
Sprint 3: RBAC Module	2 weeks	Role editor, permission matrix, assignment	Sprint 1, 2	Custom roles work
Sprint 4: Tenant Mgmt	2 weeks	Tenant CRUD, config, quotas (MSSB)	Sprint 1, 2	Tenant isolation verified
Sprint 5: Config Module	2 weeks	Settings pages, feature flags, validation	Sprint 1	Config changes audit logged
Sprint 6: Licensing UI	1 week	License viewer, usage dashboard	Sprint 1, Billing API	Usage data accurate
Sprint 7: Audit & Reports	2 weeks	Log viewer, search, export, compliance	Sprint 6	All actions logged
Sprint 8: Polish & QA	2 weeks	UX refinements, accessibility, testing	All sprints	QA sign-off obtained

Table 10.1: Admin Interface Sprint Plan

10.2 Technical Requirements

Area	Requirement	Rationale	Verification
Performance	Page load <3 seconds, interactions <200ms	User productivity	Lighthouse audit
Accessibility	WCAG 2.1 AA compliance minimum	Legal requirement, inclusivity	Automated + manual testing
Browser Support	Chrome 90+, Firefox 88+, Safari 14+, Edge 90+	Enterprise compatibility	BrowserStack testing
Responsive	Desktop (1920+), Tablet (1024+), Mobile (375+)	24/7 all emergency access	Device testing
i18n Ready	String externalization, RTL support structure	Global market readiness	Code review checklist
Error Handling	Graceful degradation, clear error messages	User trust, support efficiency	Error scenario testing
Security	XSS/CSRF protection, CSP headers, input sanitization	Platform integrity	Security review pen test

Table 10.2: Non-Functional Technical Requirements

This Admin Interface Specification provides the comprehensive blueprint for implementing the central administrative control plane for the CyberSOC Platform. The modular design enables phased delivery aligned with Go-Live roadmap priorities, starting with foundational user and role management capabilities that unblock other phases. Close integration with existing platform modules ensures consistency while the audit framework provides the compliance foundation required for enterprise and government deployments. Regular review against emerging requirements will ensure the Admin

Interface evolves alongside platform capabilities.